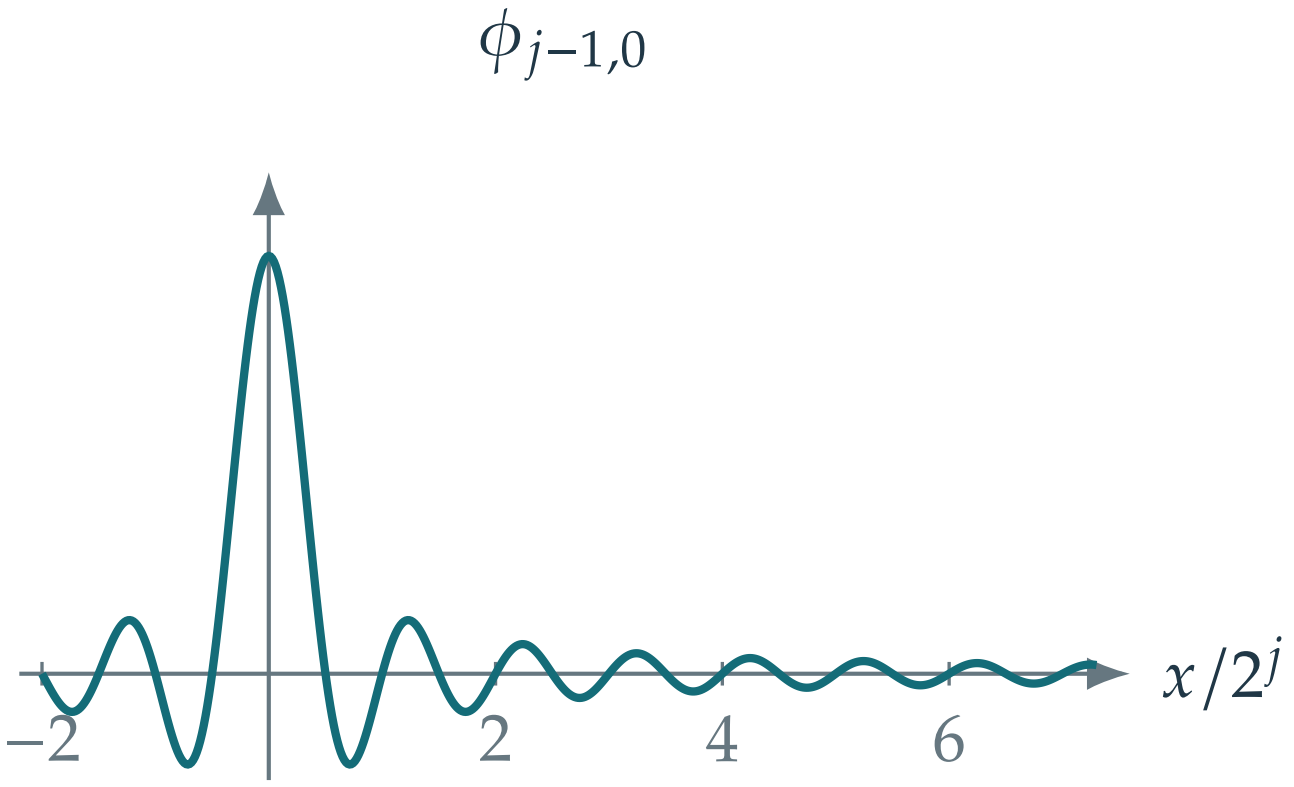

center 0, scale 2^{j-1}
relative peak $\sqrt{2}$